MODULE 42 · EXAM STRATEGY - ALL DOMAINS

Practice exam 2

Twenty more questions, on topics set 1 did not cover. Pause and answer before each reveal.

MODULE 42 of 43 DOMAIN Practice WEIGHT 20 questions

admin@sthithapragna.com

Before you start

- **Pause at each question.** Two minutes, as module 39 recommends for the beta pacing.
- **Do not look back at set 1.** These are different topics, not harder versions of the same ones.
- Read the ask, then the scenario, then the options. Every time, including the easy ones.
- **Two questions here are multiple response.** The stem tells you how many to select.
- Keep a scoresheet. You will combine it with set 1 at the end.

> ON THE EXAM
>
> Practising the working order on easy questions is what makes it automatic on the hard ones.

A team must transform 3 TB of nested JSON event logs into a flat tabular format each night. The transformation involves several joins against reference tables and a window function over user sessions.

The team has no Spark expertise and does not want to manage a cluster.

Which service should they use?

A AWS Glue, with a Spark ETL job

B AWS Lambda, invoked per file

C Amazon EMR with a long-running cluster

D Amazon Athena CTAS queries only

◆ PAUSE AND DECIDE

Which service should they use?

A AWS Glue, with a Spark ETL job

Serverless Spark with no cluster to manage, and it handles joins, window functions and nested JSON at this volume.

B AWS Lambda, invoked per file

3 TB with joins against reference tables exceeds what a 15-minute, memory-bounded function can do. Joins across files are the harder problem.

C Amazon EMR with a long-running cluster

The most capable option and the most to operate. The scenario rules it out twice: no Spark expertise, no cluster management.

D Amazon Athena CTAS queries only

Can flatten and join, and it is a query service rather than a scheduled transformation pipeline. It also struggles with complex nesting.

THE SIGNAL

Two constraints are stated plainly: no Spark expertise and no cluster. Both point away from EMR and toward the serverless option.

A data scientist fills missing values in a numeric column using the mean of the entire dataset, then splits into training, validation and test sets. Validation performance is strong; production performance is noticeably worse.

Roughly 12 percent of that column is missing.

What is the problem?

A. Mean imputation is never appropriate for numeric features

B. 12 percent missing is too high to impute at all

C. The mean was computed across all splits, so test information leaked into training

D. Missing values should have been left as nulls for the algorithm to handle

◆ PAUSE AND DECIDE

What is the problem?

A Mean imputation is never appropriate for numeric features

Mean imputation is a reasonable default for many numeric columns. The technique is not the fault here.

B 12 percent missing is too high to impute at all

12 percent is entirely workable. Very high missingness might argue for dropping a column, and that is not the described failure.

C The mean was computed across all splits, so test information leaked into training

The imputed value carries information from the validation and test rows, so those splits are no longer independent and their scores are optimistic.

D Missing values should have been left as nulls for the algorithm to handle

Some algorithms handle nulls and many do not. It is an alternative design, not an explanation of the validation-to-production gap.

THE SIGNAL

Any statistic computed before the split leaks. Fit the imputer on training only, then apply it to the other splits.

A team is choosing an embedding model for a knowledge base of 400,000 legal documents. They are weighing a 1,024-dimension model against a 256-dimension variant of the same family, which retains roughly 97 percent of retrieval accuracy.

Storage and query cost are a stated concern. The corpus takes about nine hours to embed.

What should they weigh most heavily?

A The 1,024-dimension model, since accuracy always outranks cost

B That the choice is effectively irreversible without re-embedding the whole corpus

C The generating model's context window

D The chunk overlap percentage

◆ PAUSE AND DECIDE

What should they weigh most heavily?

A The 1,024-dimension model, since accuracy always outranks cost

Accuracy does not always outrank cost, and the scenario names cost as a concern. Three percent of retrieval accuracy against a four-fold storage difference is a real trade.

B That the choice is effectively irreversible without re-embedding the whole corpus

Embeddings from different models are not comparable, so changing model means re-embedding all 400,000 documents. That is what makes this a decision to get right first.

C The generating model's context window

Relevant to how many chunks you can pass, and independent of which embedding model indexed them.

D The chunk overlap percentage

A separate tuning decision. It does not bear on which embedding model to choose.

THE SIGNAL

Nine hours to embed is the planted number. It tells you the cost of getting this wrong, not the cost of doing it.

> A lending team must check, before any model is trained, whether their historical dataset favours one demographic group over another in the labels themselves.
>
> No model exists yet. They want the check to run as a step in the data pipeline.

Which metrics apply?

A Post-training metrics, using the model's predictions

B Feature attribution drift against a baseline

C Equal opportunity difference, which compares true positive rates

D Pre-training metrics, such as class imbalance and difference in positive proportions in labels

◆ PAUSE AND DECIDE

Which metrics apply?

A Post-training metrics, using the model's predictions

Post-training metrics are functions of per-group confusion-matrix counts, which require predictions. No model exists.

B Feature attribution drift against a baseline

A monitoring concept comparing a deployed model's attributions over time. There is nothing deployed.

C Equal opportunity difference, which compares true positive rates

A true positive rate needs predictions and labels together. It is a post-training metric wearing a specific name.

D Pre-training metrics, such as class imbalance and difference in positive proportions in labels

Computed from labels and group membership alone, with no model required, so they run as a pipeline step before training.

THE SIGNAL

'No model exists yet' is the whole question. Pre-training metrics need only labels; every post-training metric needs predictions.

A team computes a customer's rolling spend average in a Spark job for training, and reimplements the same calculation in the application's Java code for real-time inference. Predictions in production differ from what offline testing predicted.

Both implementations were written by different people from the same specification.

Which TWO approaches address this?

A Increase the size of the training dataset

B Use a SageMaker AI Feature Store feature group read by both training and inference

C Retrain the model more frequently

D Use an inference pipeline so the same transformation container runs at training and serving

◆ PAUSE AND DECIDE

Which TWO approaches address this?

A Increase the size of the training dataset

More data does not reconcile two different implementations of the same feature. The model still sees different values at serving time.

B Use a SageMaker AI Feature Store feature group read by both training and inference

One computation, one stored value, read by both paths. The two implementations collapse into one, which removes the skew at source.

C Retrain the model more frequently

Retraining on the Spark version teaches the model the Spark definition again. The serving path still computes something different.

D Use an inference pipeline so the same transformation container runs at training and serving

The same container performs the transformation in both places, so there is no second implementation to diverge.

THE SIGNAL

Two implementations of one definition is training-and-serving skew. Both correct answers remove the second implementation rather than reconcile it.

A team plans to fine-tune a foundation model so it produces support replies in the company's house format and tone. They have 60,000 historical tickets of mixed quality, including many replies that were later corrected by a supervisor.

They are deciding how much data to use.

What should they do?

A Curate a few thousand high-quality, consistently formatted examples

B Use all 60,000 tickets, since more data always improves fine-tuning

C Use the corrected replies only as negative examples

D Skip fine-tuning and put the house format in the system prompt permanently

◆ PAUSE AND DECIDE

What should they do?

A Curate a few thousand high-quality, consistently formatted examples

Fine-tuning teaches the pattern it is shown. A smaller consistent set teaches the house format; a large inconsistent one teaches inconsistency.

B Use all 60,000 tickets, since more data always improves fine-tuning

The scenario states the data is mixed quality with replies that were later corrected. Volume is not the constraint; consistency is.

C Use the corrected replies only as negative examples

Fine-tuning datasets are demonstrations of desired behaviour. There is no negative-example mechanism in a standard fine-tuning format.

D Skip fine-tuning and put the house format in the system prompt permanently

A legitimate first step, and format and tone is exactly what fine-tuning is for. The scenario says they have decided to fine-tune.

THE SIGNAL

'Mixed quality, including replies later corrected' is the planted detail. It makes 60,000 a liability rather than an asset.

A security team wants to detect when a user account signs in from an IP address that is unusual for that specific account, across 90,000 accounts. They have no labelled examples of compromise.

They want a built-in SageMaker AI algorithm.

Which algorithm fits?

A XGBoost, trained on historical sign-in features

B Random Cut Forest

C IP Insights

D K-means clustering on IP address ranges

◆ PAUSE AND DECIDE

Which algorithm fits?

A XGBoost, trained on historical sign-in features

Supervised, so it needs labelled compromises. The scenario states there are none.

B Random Cut Forest

A general unsupervised anomaly detector over numeric features. It has no notion of which IP belongs to which account.

C IP Insights

Purpose-built for learning usage patterns of IPv4 addresses against entities such as account IDs, and flagging unusual entity-and-IP pairings. Unsupervised.

D K-means clustering on IP address ranges

Groups similar IPs. It does not model the relationship between a specific account and the addresses it normally uses.

THE SIGNAL

'Unusual for that specific account' is the phrase. That entity-and-IP pairing is exactly what IP Insights models.

A team must train a language model whose parameters do not fit in the memory of a single GPU. Their dataset is large but manageable, and they have a multi-GPU instance available.

Training currently fails with an out-of-memory error at model initialisation, before any batch is processed.

What should they use?

A Data parallelism, splitting batches across GPUs

B A smaller batch size

C Gradient accumulation across steps

D Model parallelism, splitting the model across GPUs

◆ PAUSE AND DECIDE

What should they use?

A Data parallelism, splitting batches across GPUs

Data parallelism puts a full copy of the model on every GPU. If one copy does not fit, neither does any of them.

B A smaller batch size

Reduces activation memory, and the failure is at initialisation before any batch is processed. Batch size is not the constraint.

C Gradient accumulation across steps

Simulates a larger batch with less memory. Same objection: the model itself does not fit.

D Model parallelism, splitting the model across GPUs

Partitions the model itself across devices, so no single GPU holds all the parameters. That is the only option that addresses the stated failure.

THE SIGNAL

'Fails at initialisation, before any batch is processed' rules out every batch-related fix in one sentence.

A team needs a model to answer questions about contracts averaging 80 pages, reading the whole document in one request. Some contracts include scanned signature pages and diagrams that must be interpreted.

Latency tolerance is generous. The workload is in a Region with limited model availability.

What should drive the model selection?

A Parameter count, as a proxy for capability

B Cost per output token

C Context window, modality support, and Regional availability

D Fine-tuning support

◆ PAUSE AND DECIDE

What should drive the model selection?

A Parameter count, as a proxy for capability

Often not published, and it does not tell you whether a model can accept 80 pages or read a diagram.

B Cost per output token

Relevant, and the answers here are short while the inputs are enormous. Input handling is the binding constraint, not output cost.

C Context window, modality support, and Regional availability

Eighty pages sets a context floor, scanned pages and diagrams require multimodal input, and limited Regional availability can eliminate a model outright.

D Fine-tuning support

Nothing in the scenario calls for changing model behaviour. The task is reading and answering.

THE SIGNAL

Three constraints are stated and all three are hard: length, modality and availability. Any one of them can eliminate a model entirely.

> An assistant answers correctly but writes in a chatty style, uses inconsistent heading formats, and ignores the company's required disclaimer block. Its factual accuracy is not in question.
>
> The team has 900 well-formatted examples of the desired output.

What should they do?

A Build a knowledge base over the style guide

B Fine-tune the model on the 900 formatted examples

C Increase the number of retrieved chunks

D Lower the temperature to zero

◆ PAUSE AND DECIDE

What should they do?

A Build a knowledge base over the style guide

Retrieval supplies knowledge. Putting a style guide in context asks the model to follow instructions it is already failing to follow.

B Fine-tune the model on the 900 formatted examples

The complaint is behaviour, not knowledge, and they have consistent demonstrations of the desired behaviour. That is fine-tuning's exact case.

C Increase the number of retrieved chunks

More context does not change output format, and it costs more tokens per request.

D Lower the temperature to zero

Makes output more deterministic, not more correctly formatted. It will chattily produce the same format every time.

THE SIGNAL

Accuracy is explicitly not in question, and the problem is format and tone. That is the clean behaviour-versus-knowledge split.

> A RAG evaluation shows faithfulness at 0.91 and answer correctness at 0.62. Context relevance is high. Reviewers report that answers are well supported by the passages provided but frequently miss part of what was asked.
>
> Retrieval returns five chunks per query.

What does this pattern indicate?

A The generating model is hallucinating and should be replaced

B Chunk size is too large, diluting relevance

C The judge model is miscalibrated

D Context coverage is the weakness: retrieval is missing part of the answer

◆ PAUSE AND DECIDE

What does this pattern indicate?

A The generating model is hallucinating and should be replaced

Faithfulness of 0.91 says the opposite: answers are well grounded in what was retrieved. Hallucination would depress faithfulness.

B Chunk size is too large, diluting relevance

Context relevance is high, so what comes back is on topic. Dilution would show as low relevance.

C Judge miscalibration

Possible in principle, and the reviewers independently report the same thing. Two sources agreeing argues against a measurement artefact.

D Context coverage is the weakness: retrieval is missing part of the answer

High faithfulness with low correctness means the model answered well from incomplete material. Relevance high and coverage low is precisely that.

THE SIGNAL

High faithfulness with low correctness is the signature: the model used what it was given faithfully, and it was not given enough.

> An internal expenses tool calls a model to categorise receipts. It receives roughly 300 requests on a working day, clustered around month end, and none overnight or at weekends. Payloads are under 200 KB and inference takes about 900 milliseconds.
>
> Users accept an occasional slower first response after a quiet period.

Which inference option is MOST cost-effective?

A. A real-time endpoint on a small instance, always on

B. Serverless inference

C. Asynchronous inference

D. Batch transform, run hourly

◆ PAUSE AND DECIDE

Which inference option is MOST cost-effective?

A. A real-time endpoint on a small instance, always on

Bills 168 hours a week to serve traffic on about 40 of them, and nothing overnight or at weekends. That is the idle endpoint problem.

B. Serverless inference

Scales to zero between requests, well inside the 4 MB and 60-second limits, and the scenario explicitly accepts cold starts.

C. Asynchronous inference

Also scales to zero, and it is built for large payloads and long processing. 200 KB and 900 ms need neither, and it adds queueing complexity.

D. Batch transform, run hourly

Users submit receipts and expect a category. Hourly batching turns an interactive tool into a waiting game.

THE SIGNAL

'Users accept an occasional slower first response' is the permission slip for cold starts, which is the only real objection to serverless.

A team hosts four large language models on one endpoint using inference components. Two of them are called only during business hours and sit idle overnight, and the team wants to stop paying for them when unused.

Requests arriving while a component is scaled to zero must not fail.

What should they do?

A Set MinInstanceCount to zero on those two components and accept the behaviour

B Move the two models to a multi-model endpoint

C Scale those two components to zero, and queue requests through asynchronous inference

D Delete the two components nightly and recreate them each morning

◆ PAUSE AND DECIDE

What should they do?

A Set MinInstanceCount to zero on those two components and accept the behaviour

Scale to zero is available, and requests arriving while scaled to zero FAIL rather than queue. The scenario forbids exactly that.

B Move the two models to a multi-model endpoint

Multi-model shares one container and one framework, and these are large language models with their own resource needs. It also still runs an instance.

C Scale those two components to zero, and queue requests through asynchronous inference

Async queues the request instead of failing it, so waking the component costs latency rather than an error. That satisfies both requirements.

D Delete the two components nightly and recreate them each morning

Achieves the saving with a fragile operational ritual, and any off-hours request fails outright.

THE SIGNAL

'Must not fail' is the constraint that rules out bare scale-to-zero, because a request arriving at zero capacity does fail.

A bank must validate a replacement fraud model against real production traffic before any customer is exposed to it. Regulation prohibits a customer decision being influenced by an unvalidated model.

They want at least two weeks of real traffic comparison.

What should they use?

A Shadow testing, replicating traffic to the new model with responses discarded

B Canary deployment with a 5 percent canary

C An A/B test with a 50/50 split

D Linear traffic shifting in 10 percent increments

◆ PAUSE AND DECIDE

What should they use?

A Shadow testing, replicating traffic to the new model with responses discarded

Traffic is replicated to the shadow variant and its responses are logged and discarded, so no customer decision is ever influenced by it.

B Canary deployment with a 5 percent canary

5 percent of customers receive decisions from the unvalidated model. Small is not zero, and the regulation says zero.

C An A/B test with a 50/50 split

Half of all customers receive the unvalidated model's decisions. It is the furthest from the requirement.

D Linear traffic shifting in 10 percent increments

Still exposes customers from the first increment. Gradual exposure is still exposure.

THE SIGNAL

'No customer decision influenced' eliminates every traffic-shifting mode at once, because all of them send real traffic to the new model.

A retailer's assistant has run on Bedrock on-demand for nine months. Usage is steady at roughly 40 million tokens a day with little variation, and finance has approved a commitment. They also want guaranteed throughput ahead of a product launch.

They use a base model with no customisation.

Which TWO statements should inform the decision?

A Provisioned Throughput reserves capacity and bills whether or not it is used

B Provisioned Throughput reduces the per-token rate on all traffic automatically

C Custom models always require Provisioned Throughput, so this is mandatory

D Nine months of steady usage gives a defensible baseline to size a commitment against

◆ PAUSE AND DECIDE

Which TWO statements should inform the decision?

A Provisioned Throughput reserves capacity and bills whether or not it is used

True, and it is the central risk. Reserved capacity is paid for continuously, so over-committing is money lost.

B Provisioned Throughput reduces the per-token rate on all traffic automatically

It reserves throughput rather than discounting on-demand traffic. Usage above the reservation does not get the reserved rate.

C Custom models always require Provisioned Throughput, so this is mandatory

Outdated, and the scenario says they use a base model anyway. Custom on-demand inference now exists for some model families.

D Nine months of steady usage gives a defensible baseline to size a commitment against

True, and it is what makes a commitment reasonable here rather than speculative. Steady measured usage is the precondition.

THE SIGNAL

The two true statements are the risk and the precondition. C is a fact that used to be true, which is the most dangerous kind of distractor.

> A team must build a process that, for each incoming insurance claim, extracts fields from the PDF, validates them against a policy database, calculates a settlement figure using a fixed formula, and emails the claimant. The sequence never varies and every branch is known in advance.
>
> An engineer proposes building this as an autonomous agent with tools.

How should you respond?

A Build it as a workflow. The sequence is known, so an agent adds cost and unpredictability

B Agree, since agents handle document extraction better than workflows

C Agree, but cap the agent at three steps to control cost

D Build it as an agent, but remove the email tool for safety

◆ PAUSE AND DECIDE

How should you respond?

A Build it as a workflow. The sequence is known, so an agent adds cost and unpredictability

If you can draw the flowchart, build the flowchart. A known invariant sequence needs orchestration, not a model deciding what to do next.

B Agree, since agents handle document extraction better than workflows

Extraction is done by a tool or service either way. The agent adds a model deciding to call it, not better extraction.

C Agree, but cap the agent at three steps to control cost

Caps the symptom. A three-step cap on a four-step fixed sequence would also break it.

D Build it as an agent, but remove the email tool for safety

Removes the step that completes the process, and still applies an agent to a fixed sequence.

THE SIGNAL

'The sequence never varies and every branch is known in advance' is the definition of a workflow. It is stated outright.

> An upstream team changes a currency field from pounds to pence without telling anyone. A deployed pricing model begins producing implausible outputs within hours. Ground truth for pricing decisions arrives 60 days later.
>
> The team wants to be told on the day this kind of change happens, not two months afterwards.

Which monitoring should they rely on?

A Model quality monitoring, comparing predictions against labels

B Data quality monitoring, comparing inputs against a baseline

C Feature attribution drift monitoring

D Scheduled evaluation against a frozen test set

◆ PAUSE AND DECIDE

Which monitoring should they rely on?

A. Model quality monitoring, comparing predictions against labels

Requires ground truth, which arrives 60 days later. It would report the problem two months after it started.

B. Data quality monitoring, comparing inputs against a baseline

Needs no labels. A hundred-fold shift in a numeric field's distribution violates the baseline immediately, which is the same-day alert they asked for.

C. Feature attribution drift monitoring

Tracks what the model relies on over time, and it also needs the model's own attributions computed on live traffic. It is slower and indirect for a unit change.

D. Scheduled evaluation against a frozen test set

Evaluates the model on unchanging data. A frozen test set cannot reveal that live inputs changed units.

THE SIGNAL

Sixty-day label latency is the planted constraint. It eliminates every label-dependent monitor and leaves the one that needs no labels.

A production endpoint serving a regulated model was reconfigured overnight to use a different model version. Nobody on the team recalls doing it. Compliance asks when the change was made, by which principal, and from where.

CloudWatch shows the endpoint behaving normally after the change.

Where should they look?

A CloudWatch Logs for the endpoint container

B Model Monitor's data quality reports

C AWS CloudTrail, for the UpdateEndpoint API call and its identity information

D The SageMaker AI Model Registry audit tab

◆ PAUSE AND DECIDE

Where should they look?

A CloudWatch Logs for the endpoint container

Container output tells you what the model did, not who called an API to change the endpoint.

B Model Monitor's data quality reports

Compares live inputs against a baseline. It has no record of configuration changes or who made them.

C AWS CloudTrail, for the UpdateEndpoint API call and its identity information

CloudTrail records console actions and API calls with identity information, which answers when, by which principal, and from where.

D The SageMaker AI Model Registry audit tab

The registry records model versions and approvals. It does not record who changed a deployed endpoint's configuration.

THE SIGNAL

'Who did it' is always CloudTrail. CloudWatch answers what the system did; CloudTrail answers who acted on it.

A company runs three generative applications: one on Bedrock, one on an open model hosted on a SageMaker AI endpoint, and one calling an external vendor API. A new policy requires identical PII redaction and topic restrictions across all three, managed centrally.

They do not want three implementations to maintain.

What should they implement?

A Bedrock Guardrails on the Bedrock application, and regex redaction in the other two

B Migrate all three applications onto Bedrock

C Amazon Macie scanning the logs of all three applications

D One guardrail, called from all three applications through the ApplyGuardrail API

◆ PAUSE AND DECIDE

What should they implement?

A Bedrock Guardrails on the Bedrock application, and regex redaction in the other two

Two implementations that will diverge from the first, and it fails the stated requirement of managing the policy centrally.

B Migrate all three applications onto Bedrock

A large migration to obtain something ApplyGuardrail provides without moving anything.

C Amazon Macie scanning the logs of all three applications

Macie classifies data at rest after the fact. It cannot redact a response before a user sees it.

D One guardrail, called from all three applications through the ApplyGuardrail API

ApplyGuardrail evaluates content independently of the model, covering Bedrock, self-hosted and third-party alike, from one configuration.

THE SIGNAL

A model outside Bedrock plus a requirement for one consistent policy is the ApplyGuardrail signal, and both halves are stated.

> A production service running on Lambda needs to call Bedrock. The security team requires that no persistent credential exists for any workload, and that access expires automatically.
>
> A developer proposes generating a long-term Bedrock API key and storing it in Secrets Manager with 90-day rotation.

How should you respond?

A Reject it. Give the Lambda function an IAM role with bedrock:InvokeModel and use no key

B Accept it, since Secrets Manager encrypts the key at rest

C Accept it, but shorten rotation to 30 days

D Reject it, and use a short-term key refreshed by the function every 12 hours

◆ PAUSE AND DECIDE

How should you respond?

A Reject it. Give the Lambda function an IAM role with bedrock:InvokeModel and use no key

A Lambda function assumes a role and receives temporary credentials automatically. No key, no persistent identity, nothing to rotate.

B Accept it, since Secrets Manager encrypts the key at rest

Storing a credential safely does not stop it being persistent, and a long-term key creates an IAM user that outlives the project.

C Accept it, but shorten rotation to 30 days

Rotates a credential that did not need to exist, and leaves the phantom IAM user in place either way.

D Reject it, and use a short-term key refreshed by the function every 12 hours

Better than a long-term key, and it builds a refresh mechanism to reproduce what an execution role already provides for free.

THE SIGNAL

API keys exist for callers that cannot assume a role. A Lambda function can, so the question of which key type never arises.

After this set

- Compare with set 1. A similar score in both suggests steady coverage.
- **A large gap between the sets** usually means one set happened to hit your weak domain.
- Combine both scoresheets and look for the domain with the most misses.
- Classify again: knowledge error, or reading error?
- Set 3 is the hardest of the three, and it is the one to sit closest to your exam date.

> ON THE EXAM
>
> Forty questions across two sets is enough to see a pattern. One set is not.